

$$1) V_B = V_{CC} \frac{R_{B2}}{R_{B1} + R_{B2}} = 3.27 \text{ V}$$

$$R_B = R_{B1} \parallel R_{B2} = 222.18 \text{ k}\Omega$$

$$I_{R_{D2}} = \frac{V_{CC} - V_O}{R_{D2}} = 0.735 \text{ mA}$$

$$V_B = R_B I_B + V_{BE} + R_{E1} (I_{E1} + I_{D2}) \quad I_{D2} = I_{R_{D2}} - I_D$$

$$V_G = V_{CE} + V_{E1}$$

$$I_D = I_{DSS} \left(\frac{V_{GS}}{V_P} - 1 \right)^2 \rightarrow I_D =$$

$$V_{GS} = V_G - R_{S2} I_D$$

$$\begin{cases} V_B = R_B I_B + V_{BE} + R_{E1} (I_{E1} + I_{D2}) \\ R_{E1} (I_{E1} + I_{D2}) = R_{D2} I_{D2} + V_O \\ I_{E1} = I_B (1 + \beta_F) \end{cases}$$

$$\rightarrow V_{BE} + R_{E1} (I_{E1} + I_{D2}) = R_{D2} I_{D2} + V_O \rightarrow I_{D2} = \frac{V_O + R_{E1} I_{E1}}{R_{E1} + R_{D2}}$$

$$V_B = R_B I_B + V_{BE} + R_{E1} I_B (1 + \beta_F) + R_{E1} \frac{R_{E1} I_B (1 + \beta_F) - V_O}{R_{E1} + R_{D2}}$$

$$I_B = \frac{V_B - V_{BE} + R_{E1} \frac{V_O}{R_{E1} + R_{D2}}}{R_B + R_{E1} (1 + \beta_F) + R_{E1} \frac{R_{E1}}{R_{E1} + R_{D2}} (1 + \beta_F)} = 6.79 \text{ }\mu\text{A}$$

$$I_C = \beta_F I_B = 1.358 \text{ mA}$$

$$I_{D2} = -303 \text{ }\mu\text{A}$$

$$I_D = I_{D2} + I_{D1} = 0.432 \text{ mA} \quad V_{CE} = V_{CC} - R_{C1} I_C - V_{E1} = 3.85 \text{ V}$$

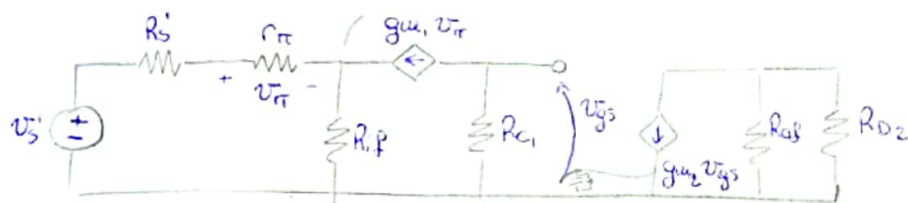
$$V_{E1} = R_{E1} (I_{E1} - I_{D2}) = 1.668 \text{ V}$$

$$I_D = I_{DSS} \left(\frac{V_{GS}}{V_P} - 1 \right)^2 \rightarrow \frac{I_D}{I_{DSS}} = \frac{V_{GS}^2}{V_P^2} + 1 - \frac{2V_{GS}}{V_P} \rightarrow V_{GS} = 3.31 \text{ V}$$

$$V_G = V_{GS} + R_{S2} I_D \rightarrow R_{S2} = \frac{V_G - V_{GS}}{I_D} = 5.11 \text{ k}\Omega$$

$$V_G = V_{CE} + V_{E1} = 5.518 \text{ V}$$

$$V_{GS} = V_O - R_{S2} I_D = 2.79 \text{ V}$$



$$A_1 = -g_{m2} R_{of} \parallel R_{D2} = -2.85$$

$$A_2 = \frac{-\beta_o R_{C1}}{R_s' + r_{\pi} + R_{if} (1 + \beta_o)} = -3.49 \quad A_{OL} = 9.95$$

$$R_{im} = R_s' + r_{\pi} + R_{if} (1 + \beta_o) = 189.08 \text{ k}\Omega$$

$$R_{out} = R_{of} \parallel R_{D2} = 4.34 \text{ k}\Omega$$

$$A_F = \frac{A_{OL}}{1 + \beta_o A_{OL}} = 5.44 \rightarrow A_V = A_F \quad \frac{R_B}{R_B + R_{S2}} = 5.42$$

$$R_{im}^F = R_{im} (1 + \beta_o A_{OL}) = 345.91 \text{ k}\Omega$$

$$R_{im} = [(R_{im}^F - R_s') \parallel R_B] + R_s = 136.13 \text{ k}\Omega$$

$$R_{out}^F = \frac{R_{out}}{1 + \beta_o A_{OL}} = 2.37 \text{ k}\Omega$$

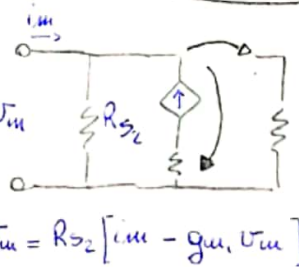
$$R_{s2}^x = (R_{im} - R_s) + R_s = R_{im}$$

$$f_L^1 = \frac{1}{2\pi R_{s2}^x C_s} = 11.69 \text{ Hz}$$

$$f_L^{2ol} = \frac{1}{2\pi R_{s2}^{xol} C_{s2}} = 13.57 \text{ Hz}$$

$$f_L^2 = \frac{f_L^{2ol}}{1 + \beta_o A_{OL}} = 7.42 \text{ Hz}$$

$$\rightarrow f_L = f_L^1 + f_L^2 = 19.11 \text{ Hz}$$



$$v_o = R_{S2} [i_{in} - g_{m1} v_{in}]$$

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$$\textcircled{2} \quad A_v(s) = 1 + \frac{R_2}{R_1} \quad \beta = \frac{R_1}{R_1 + R_2} \quad \omega_{cr} = \sqrt{A_{do} \omega_0 \omega_1} \beta = 54.23 \text{ R rad/s}$$

$$PH = 180 - \text{atan} \frac{\omega_{cr}}{\omega_0} - \text{atan} \frac{\omega_{cr}}{\omega_1} = 20.35^\circ$$

$$\omega_z = \frac{1}{C_c R_c} \quad \omega_p = \frac{1}{C_c (R_c + R_1 \parallel R_2)} \quad \omega_z = 1.38 \text{ R rad/s} \quad \omega_p = 1.36 \text{ R rad/s}$$

$$\beta_{oo}^c = \frac{R_1 \parallel R_c}{R_1 \parallel R_c + R_2} = 3.23 \cdot 10^{-4} \quad \omega_{cr}^c = A_{do} \omega_0 \beta_{oo}^c = 3.23 \text{ R rad/s}$$

$$PH = 180 - \text{atan} \frac{\omega_{cr}^c}{\omega_0} - \text{atan} \frac{\omega_{cr}^c}{\omega_1} - \text{atan} \frac{\omega_{cr}^c}{\omega_p} + \text{atan} \frac{\omega_{cr}^c}{\omega_z} = 59.54^\circ$$

